EQUATOR NETWORK COMPLIANCE REPORTING GUIDELINE

TRIPOD+AI Checklist (BMJ 2024;385:e078378)

Transparent Reporting of a multivariable prediction model of individual prognosis or diagnosis — Artificial Intelligence

BCSE335L Healthcare Data Analytics • Final Review
Evaluator: **Dr. Maheswari S (@Dr Maheswari S)** • VIT Chennai
Compliance Status: **100% Fully Compliant (All 27 Items)**

Project: DERMA-GUARD (Multimodal Dermatology AI) **Authors:** Patel Kavya Vipulkumar & Aarav Prawin **Dataset:** PAD-UFES-20 (N=2,298; CAAE: 54611416.4.0000.5060) **GitHub:** https://github.com/KavyaPatel83/DERMA-GUARD

	TRIPOD+AI Topic	Type	BMJ 2024 Guideline Description	Section & Page	Implementation Details in DERMA-GUARD
TITLE & ABSTRACT					
1	Title Identification	D;E	Identify study as developing and evaluating a clinical AI prediction model, target population, and outcome.	Title & Page 1	Title identifies clinical problem (skin lesion triage), dataset (smartphone dermatology), and AI approach (Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral).
2	Structured Summary	D;E	Structured summary: Background, Objectives, Methods, Results (with 95% CIs), and Clinical Conclusions.	Abstract (Page 1)	249-word structured summary: Background (smartphone noise), Objectives (dynamic evidence gating), Methods (7 models, QRUC profiler, LTT bounds, conformal sets, FAISS RAG), Results (89.76% Acc [95% CI: 86.6, 92.2], 0.9985 AUROC, 100% melanoma recall, 3.1% LTT miss rate), Conclusion.
INTRODUCTION					
3a	Clinical Context & Rationale	D;E	Explain healthcare context (diagnostic/prognostic), clinical setting, and rationale for AI.	Section I (Page 1)	Details cutaneous oncology epidemiology (BCC, SCC, melanoma mortality >70% if delayed), diagnostic shortages in primary dispensaries, smartphone teleconsultation potential, and failure modes of uncalibrated deep networks.
3b	Justification for New Model	D	Where existing models exist, provide clear justification for developing a new model.	Section I, II (Page 1)	Justifies developing DERMA-GUARD: existing models suffer from silent overconfidence on corrupted photos, lack finite-sample risk bounds, and provide opaque deferral pools without clinical guidance.
4	Explicit Study Objectives	D;E	State explicit scientific and clinical research questions (RQs) and hypotheses.	Section I (Page 1)	Explicitly articulates RQ1 (dynamic evidence weighting & sufficiency gating), RQ2 (distribution-free LTT risk bounds ≤5%), and RQ3 (multimodal visual RAG case grounding & DPDP Act compliance).
METHODS & MATERIALS					
5a	Data Sources	D;E	Sources of data separately for development and evaluation datasets, rationale, and representativeness.	Section IV-A (Page 2)	Open-access PAD-UFES-20 cohort collected in outpatient dermatological assistance programs at UFES, Brazil (2017–2019). CC BY 4.0 license; Ethical Approval CAAE: 54611416.4.0000.5060.
5b	Participant Eligibility	D;E	Eligibility criteria (inclusion/exclusion) and specification of settings and clinics.	Section IV-A & Fig. 1 (p. 2)	Inclusion: patient smartphone lesion photo accompanied by 26 standardized clinical metadata covariates and verified ground-truth diagnosis. Exclusion: corrupted image files or missing reference standard. N=2,298 lesions.
6	Pre-processing & Engineering	D;E	Preprocessing steps, feature engineering, and data cleaning protocols.	Section IV-C (Page 2)	Images normalized to ImageNet statistics (224x224); numeric features scaled via RobustScaler; categorical variables one-hot encoded; GrabCut lesion background segmentation. Scalers fitted strictly on training data.
7	Outcome Definition & Timing	D;E	Definition of prediction outcomes, determination method, timing, and assessor blinding.	Section IV-A & IV-E (p. 2)	Primary: 6-class diagnosis (BCC, ACK, NEV, SEK, SCC, MEL). Secondary: binary triage (malignant vs benign). >58% biopsy-confirmed histopathology; remainder verified by consensus of 3 expert dermatologists.
8	Predictor Definition & Quality	D;E	Definition of all candidate predictors, measurement timing, and predictor data quality handling.	Section IV-A, IV-C, IV-E (p. 2)	Visual: RGB photograph. Metadata: 26 clinical features (age, sex, site, Fitzpatrick type, diameter, symptoms). Text: symptom narrative. Modality quality (Q_m) evaluated via Laplacian blur variance and completeness.
9	Assessor Blinding	D;E	Blinding of outcome assessors to predictors and blinding of predictor assessors to outcome.	Section IV-A (Page 2)	Biopsy pathologists and dermatologists evaluated clinical cases during 2017–2019 independent of AI model development. AI development conducted with strict retrospective blindness.

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10	Sample Size Determination	D;E	Sample size, dataset size, event counts, and events-per-variable considerations.	Section IV-A & Table II (p. 2)	Total N=2,298 lesions across 1,373 unique patients (Train: 1,953; Val/Calibration: 229; Held-out Test: 459). 1,089 malignant events (47.4%). Ratio exceeds 50 events per candidate feature dimension.
11	Missing Data Management	D;E	Amount of missing data per predictor, assumed mechanism, and handling methods.	Section IV-C, IV-E (p. 2)	Missingness tracked across all 26 covariates. Imputed using training-set medians. Missingness fraction directly modulates Evidence Sufficiency (S), penalizing cases with unrecorded clinical history.
12	Model Development & Tuning	D	Development techniques, handling non-linearities, hyperparameter tuning, and architectures.	Section IV-C, IV-D (Page 2)	7 systematic models (A–G): ResNet50 + Multi-Head Self-Attention ViT (Image); 5-layer Residual MetaBlock MLP (Metadata); Bio_ClinicalBERT (Text); AdamW optimizer with cosine annealing schedule (30 epochs).
13	Overfitting & Leakage Guard	D	Handling overfitting and optimism during development (internal validation, patient splits).	Section IV-B & Fig. 1 (p. 2)	Strict patient-disjoint split: Grouped by patient_id (N_patients=1,373). All scaling, PCA transformations, GrabCut bounds, and temperatures fitted exclusively on training set. Zero cross-patient leakage.
14	Evaluation Protocol	E	Evaluation methods used to assess model performance on held-out evaluation dataset.	Section IV-B, V-A (p. 2, 3)	Evaluated on held-out test split of 459 lesions from 275 unseen patients. No hyperparameter adjustments or retraining conducted on test fold.
15	Performance Metrics & Curves	D;E	Specification of metrics and plots for discrimination, calibration, and clinical utility.	Section V-A & Table III, Fig. 2	Evaluates AUROC, AUPRC, Sensitivity, Specificity, Macro F1, Accuracy, and ECE—all with Wilson score and 10,000-iteration bootstrap 95% confidence intervals; ROC, PR, and reliability curves plotted.
16	Algorithmic Fairness & Subgroups	D;E	Methods to assess class imbalance, algorithmic fairness, and equity across clinical subgroups.	Section V-B, V-E (Page 3, 4)	Class-weighted cross-entropy loss handles severe melanoma rarity (n=52). Subgroup performance disaggregated across Fitzpatrick phototypes I–II (91.2%), III–IV (88.4%), and V–VI (82.6%).
17	Risk Estimates & Output Sets	D;E	Generation of model output, risk bounds, and prediction sets.	Section II-B, II-C, IV-F (p. 1, 2)	Generates calibrated probabilities via Temperature Scaling, 95% Mondrian Conformal Prediction Sets C(x) ($Pr(y \in C(x)) \geq 0.95$), and Learn-then-Test (LTT) risk bound certifying ≤3.1% missed cancer rate.
18	Ethics, Protocol & Code Access	D;E	Ethical approval, funding, conflicts of interest, protocol registration, data and code sharing.	Declarations (Page 8)	UFES Ethics Approval CAAE: 54611416.4.0000.5060; DPDP Act 2023 & ICMR AI Guidelines compliant; open data (Mendeley DOI: 10.17632/zr7vgbcyr2.1); public code: https://github.com/KavyaPatel83/DERMA-GUARD.
PATIENT & CLINICIAN INVOLVEMENT					
19	Clinician / Public Involvement	D;E	Description of clinician and stakeholder involvement in design, conduct, or reporting.	Section I, IV-E, VI (p. 1, 2, 7)	Designed in consultation with teledermatology clinical workflows: 4-tier decision gate (ACCEPT, ARBITRATE, ACQUIRE, ESCALATE) provides actionable options instead of uninformative rejection.
RESULTS					
20	Participant Flow & Demographics	D;E	Participant flow diagram, participant characteristics, demographics, and missing data frequencies.	Section IV-A, Table II, Fig. 1	Reports complete demographics: Age (59.8 ± 15.6), Sex (52.4% female), Anatomical Site (face 38.2%), Fitzpatrick types, and breakdown across all 6 diagnostic classes. Flow diagram in Fig. 1.
21	Cohort Comparison	E	Comparison of evaluation cohort characteristics with development cohort.	Section IV-A & Table II (p. 2)	Train (N=1,953), Val (N=229), and Test (N=459) sets verified to have balanced class distributions without statistically significant demographic distribution shifts ($p > 0.15$).
22	Final Model Specification	D	Final model specification (weights, architectures, code availability).	Section IV-C, D, Declarations	Full architecture specifications (ResNet50+ViT, 5L MetaBlock MLP, Bio_ClinicalBERT); model weights policy, configs, and 230 automated unit/integration tests available on GitHub.
23	Performance Estimates & CIs	D;E	Model performance estimates with 95% confidence intervals (discrimination and calibration).	Table III, V, VI (Page 3, 4)	Model G (Tri-Modal): 89.76% Acc [95% CI: 86.6, 92.2], Sens 0.9456 [95% CI: 0.918, 0.968], Spec 0.9791, Macro AUROC 0.9985, Macro AUPRC 0.9958, ECE 0.040 [95% CI: 0.028, 0.054].
24	Model Recalibration Results	D	Results of model updating (recalibration or temperature scaling).	Section IV-F, V-A, Fig. 2	Post-hoc Temperature Scaling reduced uncalibrated ECE from 0.092 down to 0.040, aligning predicted confidence with empirical observation across all 10 reliability bins.
DISCUSSION					
25	Clinical Interpretation	D;E	Overall interpretation of main results, clinical relevance, and fairness implications.	Section VI (Page 7)	Interprets multimodal synergy; 100% melanoma recall (10/10); discusses how RAG look-alike retrieval provides tangible analogical proof and Agentic AI coordination prevents

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					silent diagnostic failure.
26	**Study Limitations**	**D;E**	Study limitations, potential biases, class imbalance, and generalizability.	**Section VI-E (Page 7)**	Acknowledges modest test melanoma count (n=10) matching regional prevalence, synthetic symptom text proxy, and representation skew in dark skin phototypes (Fitzpatrick V–VI at 5%).
27a	**Input Quality Gating**	**D**	Assessment and handling of poor-quality or unavailable input data in practice.	**Section II-A, IV-E, VI-B**	Evidence Profiler explicitly computes Quality (Q_m) and Sufficiency (S); S < 0.30 triggers mandatory deferral, and 0.30 ≤ S < 0.50 triggers active VOI test acquisition (e.g., retake photo without glare).
27b	**User Interaction & Expertise**	**D**	User interaction requirements and required user expertise to safely operate the model.	**Section VI-C, VI-D (Page 7)**	Designed for assisted teleconsultation in primary clinics: 4-tier gate routes high-risk cases to ESCALATE for mandatory board-certified dermatologist review; human-in-the-loop oversight is strictly preserved.
27c	**Next Steps & Translation**	**D;E**	Future research, clinical translation, prospective validation, and safety monitoring.	**Section VII (Page 7)**	Identifies future integration of clinical vision-language foundation models (PanDerm), prospective multi-reader multi-case (MRMC) trials, and multi-centre international evaluation.

Formal Attestation of Compliance: We hereby certify that the manuscript "*DERMA-GUARD: Evidence-Aware Specialist Consensus with Risk-Controlled Evidence Acquisition and Deferral for Multimodal Dermatology*" has been developed, evaluated, and reported in full accordance with the **TRIPOD+AI Statement (BMJ 2024;385:e078378)** as catalogued by the EQUATOR Network. All 27 items (and associated sub-items) have been implemented, verified, and mapped to the manuscript.

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